

Slot-C

Analytical problem

1. Grey level Transformation

a. Purpose of Grey level Transformation

Grey level transformation is used to improve the quality of an image by changing the brightness values (pixel values). It helps in

* Enhance image contrast.

* Highlight important details.

b. Negative Transformation

Formula:

$$S = 255 - r$$

Where:

* r = Original pixel value.

* S = Transformed pixel value.

Original pixel	Calculation	new pixel
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Transformed pixel:

205, 155, 105, 55

2. Histogram Equalization

a why Histogram Equalization is used.

Histogram Equalization Improves the contrast of an image by spreading the pixel intensity values over the entire intensity range.

Advantages

- * Improves image contrast.
- * Makes hidden details visible.
- * Enhances low-contrast images.
- * Useful in medical and satellite imaging.

b Compute the Cumulative Distribution Function:-

Intensity	Frequency
0	4
1	6
2	10
3	6

Total pixels.

$$N = 4 + 6 + 10 + 6 = 26$$

Intensity	Frequency	Cumulative Frequency
0	4	4
1	6	10
2	10	20
3	6	26

$$CDF = \frac{\text{Cumulative Frequency}}{26}$$

Intensity	CDF
0	$\frac{4}{26} = 0.154$
1	$\frac{10}{26} = 0.385$
2	$\frac{20}{26} = 0.769$
3	$\frac{26}{26} = 1.000$

Equalized intensity values.

4-bit image

$$L=16$$

Formula:-

$$S = (L-1) \times \text{CDF}$$

$$L-1 = 15$$

Intensity	CDF	Equalized value
0	0.154	$15 \times 0.154 = 2$
1	0.385	$15 \times 0.385 = 6$
2	0.769	$15 \times 0.769 = 12$
3	1.000	$15 \times 1 = 15$

Equalized intensities:-

0 $\rightarrow$ 2 1 $\rightarrow$ 6 2 $\rightarrow$ 12 3 $\rightarrow$ 15.

3. Gaussian Smoothing.

a) Why Gaussian filtering is preferred.

Gaussian filtering is preferred because:

- * Removes image noise effectively.
- * Produces smoother images.
- * Gives more importance to the center pixel.
- * Preserves edges better than an averaging filter.

b. Apply Gaussian Kernel

Gaussian Kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image Region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Each Element.

$$(1 \times 10) + (2 \times 20) + (1 \times 10)$$

$$+ (2 \times 20) + (4 \times 40) + (2 \times 20)$$

$$+ (1 \times 10) + (2 \times 20) + (1 \times 10)$$

Top Row:-

$$10 + 40 + 10 = 60$$

Middle Row.

$$40 + 160 + 40 = 240$$

Bottom Row

$$10 + 40 + 10 = 60$$

Add All Values

$$60 + 240 + 60 = 360$$

Divide by 16

$$\frac{360}{16} = 22.5$$

Round Value

23

New Central pixel = 23

4. Edge Detection Using Sobel Operator.

a. Purpose of sobel Edge Detection.

- * Identify object boundaries.
- * Highlight edges in an image.
- * Prepare images for segmentation and object recognition.

b. Sobel Calculation

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Sobel x kernel

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Sobel y kernel

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Calculate G_x .

$$\begin{aligned} & (-1 \times 10) + (0 \times 10) + (1 \times 10) \\ & + (-2 \times 10) + (0 \times 50) + (2 \times 10) \\ & + (-1 \times 10) + (0 \times 10) + (1 \times 10) \end{aligned} = 0$$

Calculate G_y

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10)$$

$$+ (0 \times 10) + (0 \times 50) + (0 \times 10)$$

$$+ (1 \times 10) + (2 \times 10) + (1 \times 10)$$

Gradient Magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{0^2 + 0^2} = 0$$

Gradient Magnitude = 0

5. Contrast Stretching:-

a. Need for Contrast Stretching.

Contrast stretching is an image enhancement technique used to improve the visibility of an image.

* increase the difference b/w dark & bright areas.

* Makes hidden details easier to see.

* improves the overall quality of low-contrast images.

b. Linear Contrast Stretching:-

input intensity range = 50 to 150

output intensity range = 0 to 255

Formula:-

$$S = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

Where

$$* r_{\min} = 50$$

$$* r_{\max} = 150$$

So,

$$S = \frac{(r-50)}{100} \times 255$$

$$r = 50$$

$$S = \frac{(50-50)}{100} \times 255$$

$$S = 0 \times 255 = 0$$

$$\text{Output} = 0$$

$$r = 100$$

$$S = \frac{(100-50)}{100} \times 255$$

$$= \frac{50}{100} \times 255$$

$$= 0.5 \times 255 = 127.5 \approx 128$$

$$\text{Output} = 128$$

$$r = 150$$

$$S = \frac{(150-50)}{100} \times 255$$

$$= \frac{100}{100} \times 255$$

$$\text{Output} = 255$$

Therefore:

50 becomes 0 (black)

100 becomes 128 (middle gray)

150 becomes 255 (white)